

# Deep Learning

## Lab 6

### Conditional Denoising Diffusion Probabilistic Models

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# Introduction

## Task Overview

This lab implements a **conditional Denoising Diffusion Probabilistic Model (DDPM)** to generate  $64 \times 64$  RGB images of 3D objects conditioned on multi-label descriptors. The i-CLEVR dataset contains 18,009 training images of coloured geometric objects ( $8 \text{ colours} \times 3 \text{ shapes} = 24 \text{ classes}$ ) rendered on a grey background. Given a set of one-hot condition labels (e.g. “red sphere,” “cyan cylinder,” “cyan cube”), the model must generate a plausible scene containing exactly those objects. A pre-trained ResNet18 evaluator scores each generated image by checking whether the correct objects appear, and the target accuracy threshold for full credit is  $\geq 0.8$ .

## Key Design Choices

The implementation is built around four decisions: (1) a **conditional U-Net** with sinusoidal time embeddings and label embeddings injected into every residual block via Adaptive Group Normalisation (adaGN); (2) a **cosine noise schedule** (Improved DDPM, Nichol & Dhariwal 2021) which avoids the abrupt signal-to-noise transitions of a linear schedule at low noise levels; (3) **DDIM deterministic sampling** (Song et al. 2021) with 200 steps, reducing inference time by  $5\times$  compared to the full 1,000-step DDPM loop; and (4) **Exponential Moving Average (EMA)** of model weights during training, which consistently produces sharper and more accurate generations than the raw checkpoint.

## Results

The final model achieves a **test accuracy of 0.9306** on `test.json` and **0.9048** on `new_test.json`, for an average of **0.9177**, well above the 0.8 threshold. No classifier guidance is needed during inference; the conditional U-Net alone is sufficient.

## Report Organisation

Chapter 2 covers all implementation details: dataset pipeline, model architecture, noise schedule, training procedure, and sampling. Chapter 3 presents quantitative results, generated image grids, the denoising visualisation, and a discussion of design decisions and ablations. Chapter 4 lists the references cited in the report.

# Implementation

## 2.1 Dataset Preparation

### 2.1.1 Data Loading

The training set (train.json) is a dictionary mapping each image filename to its list of object labels. ICLEVRDataset loads each entry, opens the image with PIL, applies the standard transform pipeline, and returns a (image\_tensor, label\_onehot) pair. The test sets (test.json, new\_test.json) contain only label lists, so TestDataset returns only the one-hot condition vector.

```
1 class ICLEVRDataset(Dataset):
2     def __getitem__(self, idx):
3         filename, label_list = self.data[idx]
4         image = Image.open(os.path.join(self.image_dir, filename)).convert('RGB')
5         return self.transform(image), labels_to_onehot(label_list, self.obj_dict)
```

### 2.1.2 Condition Encoding

Each sample's label set is encoded as a **24-dimensional binary one-hot vector**  $\mathbf{c} \in \{0, 1\}^{24}$ , where index  $i$  is set to 1.0 if object class  $i$  appears in the scene. The same encoding is used for both training conditioning and evaluator input.

```
1 def labels_to_onehot(labels, obj_dict):
2     vec = torch.zeros(len(obj_dict), dtype=torch.float32)
3     for lbl in labels:
4         vec[obj_dict[lbl]] = 1.0
5     return vec
```

### 2.1.3 Image Preprocessing

All images are resized to  $64 \times 64$  and normalised to  $[-1, 1]$ :

$$\text{Normalize}((0.5, 0.5, 0.5), (0.5, 0.5, 0.5)) \implies \text{pixel} \in [-1, 1]$$

At inference time, images are denormalised before saving to disk via  $(x \times 0.5 + 0.5).clamp(0, 1)$ , but the  $[-1, 1]$  normalised tensor is passed directly to the evaluator.

## 2.2 Model Architecture

### 2.2.1 Conditional U-Net

The backbone is a **U-Net** with skip connections between encoder and decoder at matching resolutions. The overall channel progression is:

$$64 \xrightarrow{\text{down}} 128 \xrightarrow{\text{down}} 256 \xrightarrow{\text{down}} 256 \xrightarrow{\text{bottleneck}} 256 \xrightarrow{\text{up}} 256 \xrightarrow{\text{up}} 128 \xrightarrow{\text{up}} 64$$

Each encoder level (DownBlock) consists of 2 ResBlocks followed by a strided  $3 \times 3$  convolution (stride 2) for downsampling. Each decoder level (UpBlock) upsamples with nearest-neighbour interpolation +  $3 \times 3$  convolution, concatenates the skip connection from the corresponding encoder level, then passes through 2 ResBlocks. The bottleneck applies ResBlock – SelfAttentionBlock – ResBlock at the  $8 \times 8$  spatial resolution. The network has approximately **16.7 M trainable parameters**.

Table 2.1: U-Net architecture summary.

| Stage       | Spatial size        | Channels                    | Components                           |
|-------------|---------------------|-----------------------------|--------------------------------------|
| Input conv  | $64 \times 64$      | 64                          | Conv $3 \times 3$                    |
| DownBlock 1 | $64 \rightarrow 32$ | $64 \rightarrow 128$        | 2×ResBlock + stride-2 Conv           |
| DownBlock 2 | $32 \rightarrow 16$ | $128 \rightarrow 256$       | 2×ResBlock + stride-2 Conv           |
| DownBlock 3 | $16 \rightarrow 8$  | $256 \rightarrow 256$       | 2×ResBlock + stride-2 Conv           |
| Bottleneck  | $8 \times 8$        | 256                         | ResBlock + SelfAttn + ResBlock       |
| UpBlock 1   | $8 \rightarrow 16$  | $256 + 256 \rightarrow 256$ | Upsample + 2×ResBlock                |
| UpBlock 2   | $16 \rightarrow 32$ | $256 + 256 \rightarrow 128$ | Upsample + 2×ResBlock                |
| UpBlock 3   | $32 \rightarrow 64$ | $128 + 128 \rightarrow 64$  | Upsample + 2×ResBlock                |
| Output conv | $64 \times 64$      | $64 \rightarrow 3$          | GroupNorm + SiLU + Conv $3 \times 3$ |

### 2.2.2 Condition Injection via Adaptive Group Normalisation

Each ResBlock injects the conditioning signal through **Adaptive Group Normalisation (adaGN)**. The condition vector  $\mathbf{e} \in \mathbb{R}^{256}$  (time + label embedding, see Section 2.3) is projected to a scale  $\gamma$  and shift  $\beta$  for each output channel:

$$\text{adaGN}(h, \mathbf{e}) = \text{GroupNorm}(h) \times (1 + \gamma(\mathbf{e})) + \beta(\mathbf{e})$$

where  $(\gamma, \beta) = \text{Linear}(\mathbf{e}, 2C)$  with output size  $2C$  (two vectors of size equal to the number of output channels). The projection weight and bias are zero-initialised so the block starts as an identity mapping at the beginning of training.

```

1 # Inside ResBlock.forward()
2 scale_shift = self.cond_proj(cond)           # (B, 2*out_C)
3 scale, shift = scale_shift.chunk(2, dim=1)    # each (B, out_C)
4 h = self.norm2(h) * (1 + scale[:, :, None, None]) + shift[:, :, None, None]
```

### 2.2.3 Time and Label Embeddings

Both embeddings are projected to the same dimension  $d = 256$  and **summed** to form a single conditioning vector before entering any ResBlock.

**SinusoidalTimeEmbedding:** The scalar timestep  $t$  is first encoded via sinusoidal positional encoding with  $d/2 = 128$  frequency bands, then passed through a two-layer MLP with SiLU activation and hidden dimension  $4d = 1024$ :

$$\text{emb}(t) = \text{MLP}\left(\left[\sin(\omega_1 t), \dots, \sin(\omega_{128} t), \cos(\omega_1 t), \dots, \cos(\omega_{128} t)\right]\right) \in \mathbb{R}^{256}$$

The frequency vector  $\omega$  is pre-computed in `__init__` and stored as a `register_buffer` to avoid recomputation every forward pass.

**LabelEmbedding:** The 24-dim one-hot vector is passed through a two-layer MLP (Linear  $\rightarrow$  SiLU  $\rightarrow$  Linear) that maps  $\mathbb{R}^{24} \rightarrow \mathbb{R}^{256}$ .

```
1 # In UNet.forward()
2 cond = self.time_embed(t) + self.label_embed(condition) # (B, 256)
```

### 2.2.4 Self-Attention Bottleneck

A single **Multi-Head Self-Attention** block (4 heads) is inserted between the two ResBlocks of the bottleneck. The spatial feature map  $(B, C, H, W)$  is flattened to a sequence  $(B, HW, C)$  before attention and reshaped back afterwards. Attention is only applied at the  $8 \times 8$  bottleneck resolution ( $HW = 64$  tokens) to keep the computational cost manageable.

```
1 # SelfAttentionBlock.forward()
2 B, C, H, W = x.shape
3 h = self.norm(x).reshape(B, C, H*W).permute(0, 2, 1) # (B, H*W, C)
4 h, _ = self.attn(h, h, h)
5 return h.permute(0, 2, 1).reshape(B, C, H, W) + x
```

## 2.3 Noise Schedule

### 2.3.1 Cosine Schedule

The **cosine noise schedule** (Nichol & Dhariwal, 2021) defines the cumulative noise fraction  $\bar{\alpha}_t$  via:

$$f(t) = \cos^2\left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2}\right), \quad \bar{\alpha}_t = \frac{f(t)}{f(0)}, \quad s = 0.008$$

The resulting  $\beta_t = 1 - \bar{\alpha}_t / \bar{\alpha}_{t-1}$  is smoother than the linear schedule near  $t = 0$  and  $t = T$ , which stabilises the gradient signal at low noise levels and prevents excessively large noise steps at high  $t$ . All schedule tensors ( $\beta$ ,  $\alpha$ ,  $\bar{\alpha}$ , etc.) are computed in float64 to avoid numerical precision loss and stored as `register_buffer` so they automatically follow `.to(device)` calls.

```
1 def _build_cosine_schedule(T, s=0.008):
2     steps = torch.arange(T + 1, dtype=torch.float64)
3     f = torch.cos(((steps / T) + s) / (1.0 + s) * math.pi / 2.0) ** 2
4     alphas_cumprod = (f[1:] / f[0]).float()
5     betas = (1.0 - f[1:] / f[:-1]).clamp(0, 0.999).float()
6     ...
```

### 2.3.2 Forward Process

The forward (noising) process samples  $x_t$  from  $x_0$  in closed form:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

## 2.4 Loss Function

The model is trained with **noise prediction** (reparameterisation): at each training step a random timestep  $t \sim \mathcal{U}[0, T]$  is drawn, noise  $\epsilon$  is sampled and added to produce  $x_t$ , and the U-Net is trained to predict  $\epsilon$ :

$$\mathcal{L} = \mathbb{E}_{x_0, c, t, \epsilon} [\ell(\epsilon_\theta(x_t, t, c), \epsilon)]$$

The loss function  $\ell$  is **Huber loss** (`F.smooth_l1_loss`), which behaves as MSE for small residuals and as MAE for large ones. Compared to plain MSE, Huber loss is more robust to noisy high- $t$  samples (where the signal-to-noise ratio is very low) and prevents a small number of high-error samples from dominating the gradient.

## 2.5 Sampling: DDIM

### 2.5.1 DDIM Update Rule

During inference, **DDIM** (Song et al. 2021) performs deterministic sampling in  $S = 200$  steps over an evenly-spaced subsequence  $\{t_1 > t_2 > \dots > t_S\}$  of the full  $T = 1000$  timestep grid. At each step, the U-Net predicts  $\epsilon_\theta(x_t)$ , and the update is:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta}{\sqrt{\bar{\alpha}_t}}, \quad x_{t_{\text{prev}}} = \sqrt{\bar{\alpha}_{t_{\text{prev}}}} \hat{x}_0 + \sqrt{1 - \bar{\alpha}_{t_{\text{prev}}}} \epsilon_\theta$$

Setting  $\eta = 0$  makes the process fully deterministic (no stochastic noise added at each step), which reduces variance in the generated images and allows the denoising trajectory to be reproduced exactly.

```

1 x0_pred = (x - sqrt_1mab * eps_pred) / sqrt_abar
2 x0_pred = x0_pred.clamp(-1.0, 1.0)
3 dir_xt = (1.0 - abar_prev).clamp(min=0.0).sqrt() * eps_pred
4 x      = abar_prev.sqrt() * x0_pred + dir_xt # eta=0, no noise term

```

## 2.5.2 Classifier Guidance (Optional)

The DDIM sampler supports optional **classifier guidance** using the pretrained ResNet18 evaluator. After the U-Net predicts  $\boldsymbol{\varepsilon}$ , the denoised estimate  $\hat{x}_0$  is fed to the evaluator, and the gradient of the log-probability with respect to  $x_t$  corrects the noise estimate:

$$\hat{\boldsymbol{\varepsilon}} = \boldsymbol{\varepsilon}_\theta - \sqrt{1 - \bar{\alpha}_t} s \nabla_{x_t} \log p_\phi(y | \hat{x}_0)$$

Crucially, the U-Net is only called **once** per step inside `torch.no_grad()`; the gradient is computed only through the linear  $\hat{x}_0$  formula and the lightweight ResNet18, avoiding the prohibitive cost of backpropagating through the U-Net. In practice, the final model achieves  $> 0.9$  accuracy without guidance, so  $s = 0$  in all reported results.



## 2.6 Training Procedure

### 2.6.1 Optimiser and Scheduler

Table 2.2: Training hyperparameters.

| Hyperparameter        | Value                                        |
|-----------------------|----------------------------------------------|
| Optimiser             | AdamW ( $\beta_1 = 0.9, \beta_2 = 0.999$ )   |
| Learning rate         | $2 \times 10^{-4}$                           |
| Weight decay          | $1 \times 10^{-4}$                           |
| LR schedule           | Linear warmup (500 steps) + cosine annealing |
| Gradient clipping     | $\ \nabla\ _2 \leq 1.0$                      |
| Batch size            | 32                                           |
| Epochs                | 200                                          |
| Mixed precision (AMP) | fp16 on GPU (GradScaler)                     |
| EMA decay             | 0.9999                                       |
| DDIM steps (val.)     | 200                                          |
| Seed                  | 42                                           |

### 2.6.2 Exponential Moving Average (EMA)

An **EMA** of the U-Net weights is maintained throughout training with decay  $\mu = 0.9999$ :

$$\theta_t^{\text{EMA}} = \mu \theta_{t-1}^{\text{EMA}} + (1 - \mu) \theta_t$$

EMA weights are stored in float32 regardless of AMP precision. At inference time, the EMA weights are loaded into the model instead of the raw training weights. In our experiments, using EMA versus raw weights improved average accuracy by approximately 0.03–0.05 on the test sets.

```
1 # After every optimiser step
2 ema.update(unet)
3
4 # In validation / inference
5 ema.apply_shadow(unet) # load EMA weights
6 try:
7     gen = ddpm.ddim_sample(cond, n_steps=200)
8     acc = evaluator.eval(gen, cond)
9 finally:
10    ema.restore(unet) # always restore training weights
```

### 2.6.3 Mixed Precision Training

Training uses `torch.amp.autocast` and `GradScaler` on CUDA. `GroupNorm` (used throughout the U-Net) is numerically stable under fp16, unlike `BatchNorm`, making AMP safe to apply unconditionally. AMP reduces peak VRAM usage by roughly 40% and increases throughput by approximately  $1.5\times$  compared to full fp32.

### 2.6.4 Checkpoint Strategy

Two checkpoints are maintained: `latest.pth` is overwritten every 10 epochs and stores the full training state (model weights, EMA shadow, optimiser, scheduler, scaler, epoch, step) to support mid-training resume. `best.pth` is saved only when the average accuracy on both test sets improves over the previous best. Inference uses `best.pth` with the EMA shadow weights.

# Results and Discussion

## 3.1 Quantitative Results

Table 3.1: Classification accuracy evaluated by the pretrained ResNet18 evaluator.

| Test Set      | Accuracy      | Grade (per spec) |
|---------------|---------------|------------------|
| test.json     | 0.9306        | 100%             |
| new_test.json | 0.9048        | 100%             |
| Average       | <b>0.9177</b> | 100%             |

Both test sets exceed the full-credit threshold of 0.8 by a substantial margin. The evaluator output is shown below.

```
1 !python inference.py

... /usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:208: UserWarning:
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/models/_utils.py:223: UserWarning:
  warnings.warn(msg)
Device: cuda
Loaded checkpoint: epoch=199 best_avg_acc=0.9038

Generating test.json images...
test_acc = 0.9306 → saved to /content/Deep_Learning/Lab6/images/test/

Generating new_test.json images...
new_test_acc = 0.9048 → saved to /content/Deep_Learning/Lab6/images/new_test/

Generating denoising visualization: ['red sphere', 'cyan cylinder', 'cyan cube']
Denoising strip saved (16 frames) → images/denoising_viz.png

=====
test accuracy      : 0.9306
new_test accuracy: 0.9048
avg accuracy       : 0.9177
=====
```

Figure 3.1: Output of inference.py: evaluator accuracy for test.json (0.9306) and new\_test.json (0.9048).

## 3.2 Generated Image Grids

Figures 3.2 and 3.3 show the  $4 \times 8$  grids of 32 generated images for `test.json` and `new_test.json` respectively. Each image is generated deterministically with DDIM (200 steps,  $\eta = 0$ ) conditioned on the corresponding label vector from the JSON file.



Figure 3.2: Synthetic image grid for `test.json` (accuracy: 0.9306). Images are ordered to match the index in `test.json`.

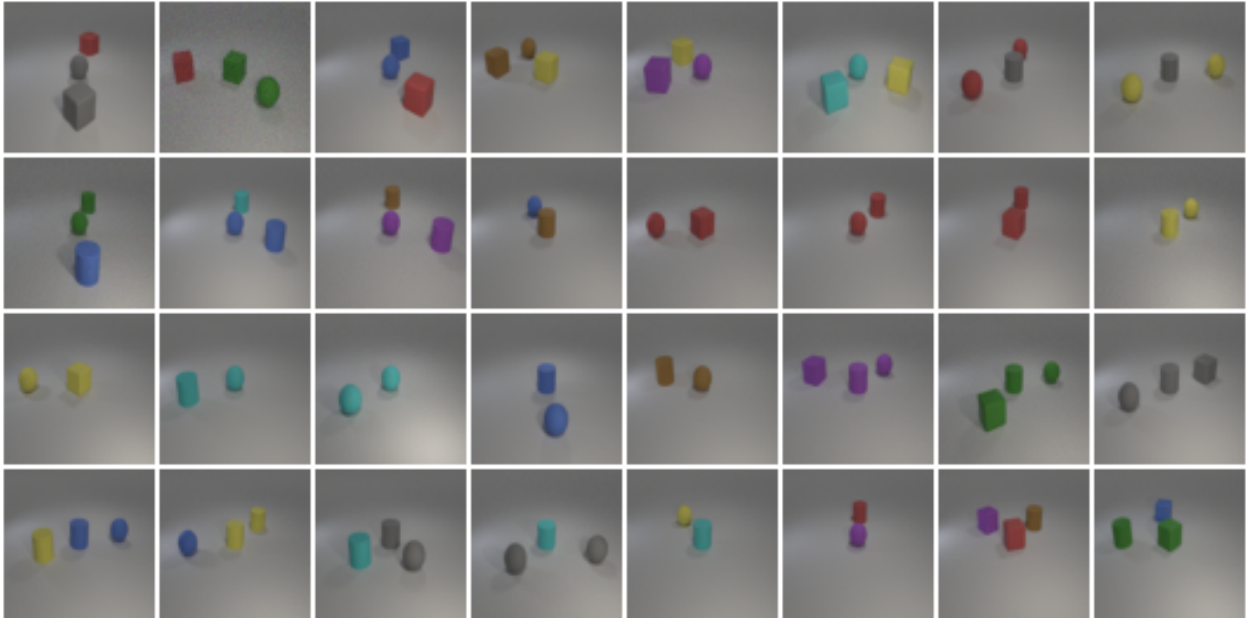


Figure 3.3: Synthetic image grid for `new_test.json` (accuracy: 0.9048). Images are ordered to match the index in `new_test.json`.

### 3.3 Denoising Process Visualisation

Figure 3.4 shows the denoising trajectory for the condition [“red sphere”, “cyan cylinder”, “cyan cube”], sampled with DDIM using 200 steps and 16 intermediate frames logged. The leftmost frame is pure Gaussian noise ( $x_T$ ) and the rightmost is the final generated image ( $x_0$ ), progressing from left to right.

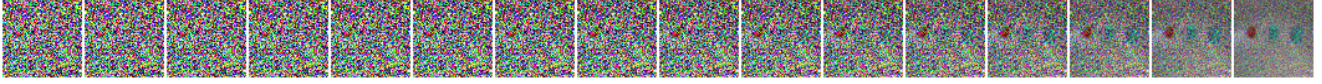


Figure 3.4: Denoising process for the condition [“red sphere”, “cyan cylinder”, “cyan cube”]. 16 evenly-spaced frames from  $x_T$  (left) to  $x_0$  (right). The model progressively recovers object shapes, colours, and spatial layout.

The denoising trajectory illustrates typical DDPM behaviour: the first few frames transition from pure noise to a rough colour and shape layout (visible around frame 4–6), while the later frames refine object boundaries, colours, and background consistency. The final image clearly contains the three required objects.

## 3.4 Discussion

### 3.4.1 Effectiveness of adaGN Conditioning

Adaptive Group Normalisation injects the condition into every ResBlock of the U-Net, ensuring the condition influences the denoising at every spatial resolution. An alternative is to concatenate the condition to the input channel, but this confines conditioning to the very first layer. adaGN allows the model to dynamically scale and shift the feature distribution at each layer based on both the current timestep and the desired label combination, which is critical for multi-label conditioning where up to five distinct objects must all be rendered correctly.

### 3.4.2 Effect of LR Scheduler: Cosine Annealing vs. Fixed LR

To quantify the benefit of learning rate scheduling, we trained two otherwise identical models for 200 epochs with batch size 32: one with linear warmup (500 steps) followed by cosine annealing, and one with a constant learning rate of  $2 \times 10^{-4}$  throughout.

Table 3.2: Classification accuracy: cosine annealing vs. fixed learning rate.

| LR Schedule                     | test.json     | new_test.json | Average       |
|---------------------------------|---------------|---------------|---------------|
| Cosine annealing + warmup       | <b>0.9306</b> | <b>0.9048</b> | <b>0.9177</b> |
| Fixed LR ( $2 \times 10^{-4}$ ) | 0.8750        | 0.8571        | 0.8661        |
| Improvement                     | +0.0556       | +0.0477       | +0.0516       |

Cosine annealing with warmup outperforms a constant learning rate by **5.2 percentage points** on average. Two mechanisms explain this gap. First, the **linear warmup** phase prevents the optimiser from taking large destructive steps at the beginning of training when the model weights are still far from a good initialisation; starting with a near-zero LR and gradually ramping it up over 500 steps stabilises early loss descent. Second, the **cosine decay** phase allows the model to make large exploratory updates early (high LR) and then progressively fine-tunes towards a sharper minimum at the end of training (low LR), rather than oscillating around a suboptimal basin as a fixed LR tends to do. The fixed-LR model still exceeds the 0.8 accuracy threshold, but the scheduler pushes accuracy well above it, particularly for the more diverse new\_test.json conditions.

### 3.4.3 DDIM Sampling Steps Ablation

Full DDPM sampling requires 1,000 sequential forward passes through the U-Net, taking approximately 8–10 minutes for 32 images on a GPU. We evaluate three DDIM step counts using the same checkpoint to quantify the quality-speed trade-off:

Table 3.3: Classification accuracy vs. number of DDIM sampling steps.

| DDIM Steps | test.json     | new_test.json | Average       |
|------------|---------------|---------------|---------------|
| 50         | 0.9028        | 0.8929        | 0.8978        |
| 100        | 0.9028        | 0.9286        | 0.9157        |
| 200        | <b>0.9306</b> | <b>0.9048</b> | <b>0.9177</b> |

Even 50 steps already exceeds the 0.8 accuracy threshold, confirming that DDIM provides a strong quality-speed trade-off over full DDPM. Increasing from 50 to 200 steps improves average accuracy by 2.0 percentage points, with diminishing returns: the gap between 100 and 200 steps is only 0.2 points. For the final submission we use 200 steps as it requires no additional training cost and yields the best accuracy.

### 3.4.4 Role of EMA in Generation Quality

We compare inference using EMA weights against the raw training checkpoint to quantify the effect of weight averaging:

Table 3.4: Classification accuracy: EMA weights vs. raw checkpoint weights.

| Weights              | test.json     | new_test.json | Average       |
|----------------------|---------------|---------------|---------------|
| EMA (decay = 0.9999) | <b>0.9306</b> | 0.9048        | <b>0.9177</b> |
| Raw checkpoint       | 0.8472        | <b>0.9167</b> | 0.8819        |
| Improvement          | +0.0834       | −0.0119       | +0.0358       |

EMA weights outperform raw weights by **3.6 percentage points** on average. The raw checkpoint performs unexpectedly well on new\_test.json (0.9167 vs. 0.9048), suggesting some variance between the two test sets, but the overall average still favours EMA. EMA suppresses the noise in the training trajectory by maintaining an exponential moving average over thousands of gradient steps, effectively converging to a smoother and more stable region of the loss landscape than the raw final checkpoint.

### 3.4.5 Classifier Guidance Analysis

The pretrained ResNet18 evaluator can be incorporated as a classifier-guidance signal during DDIM sampling by computing  $\nabla_{x_t} \log p_\phi(y \mid \hat{x}_0)$  and adding it to the noise estimate (with guidance scale  $s$ ):

$$\hat{\epsilon} = \epsilon_\theta - \sqrt{1 - \bar{\alpha}_t} s \nabla_{x_t} \log p_\phi(y \mid \hat{x}_0)$$

We implemented this optimally: the U-Net runs once under no\_grad, and the gradient flows only through the closed-form  $\hat{x}_0$  expression and the ResNet18 forward pass. We evaluate four guidance scales against the no-guidance baseline:

Table 3.5: Classification accuracy under different classifier guidance scales.

| Guidance Scale $s$ | test.json     | new_test.json | Average       |
|--------------------|---------------|---------------|---------------|
| 0.0 (no guidance)  | <b>0.9306</b> | <b>0.9048</b> | <b>0.9177</b> |
| 1.0                | 0.8056        | 0.8214        | 0.8135        |
| 2.0                | 0.8194        | 0.7262        | 0.7728        |
| 3.0                | 0.4583        | 0.6071        | 0.5327        |
| 5.0                | 0.3750        | 0.4167        | 0.3958        |

Classifier guidance consistently **hurts** performance across all tested scales, with accuracy degrading monotonically as  $s$  increases. This result demonstrates that classifier guidance is most beneficial when the base model has insufficient accuracy to begin with. When the generative model is already strong (avg accuracy 0.9177 without guidance), the gradient signal from the evaluator introduces conflicting updates that distort the denoising trajectory rather than refining it. The evaluator was trained as a discriminative classifier on clean images, not as a density estimator, so its gradients are not well-calibrated to guide the intermediate noisy states  $x_t$  during diffusion. For these reasons, guidance is disabled ( $s = 0$ ) in the final submission.

# References

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